

UNIVERSITY OF PROFESSIONAL STUDIES, ACCRA (UPSA)

DEGREE PROGRAMME

REGULAR SCHOOL



END OF SECOND SEMESTER EXAMINATIONS- 2017/2018 ACADEMIC YEAR

FACULTY OF ACCOUNTING & FINANCE

DEPARTMENT OF BANKING & FINANCE

LEVEL 200

BCPC 208: QUANTITATIVE METHODS

Date: Friday, 1st June, 2018.

TIME ALLOWED: 3 HOURS

University of Professional Studies, Accra.
P. O. Box LG 149, Accra
Graduate Library

INDEX NUMBER

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INSTRUCTION(S):

THERE ARE TWO (2) SECTIONS IN THIS PAPER, SECTIONS A & B.

SECTION A: ANSWER ALL QUESTIONS

SECTION B: ANSWER ANY THREE (3) QUESTIONS

EACH QUESTION CARRIES 15 MARKS

SECTION A

ANSWER ALL QUESTIONS IN THIS SECTION

QUESTION ONE

- a) The demand and supply equations for a good are given as $2P = 41 - 3Q_d$ and $5P = 130 + 3Q_s$, where P , Q_d and Q_s denote the price, quantity demanded, and quantity supplied, respectively. Determine how much a consumer will pay for each unit of the good if a fixed tax of ₵5 per unit is imposed on the good. (3 marks)
- b) How much money should a person save in an account paying 5% interest compounded after every six months if he or she wants to have GH₵ 8000 in 3 years? (3 marks)
- c) The sales manager of a company wishes to find out whether the average monthly sales of all their 100 outlets in Ghana, with a standard deviation of ₵2715, is different from ₵23500. She randomly visited 30 outlets of the company and computed their average monthly sales to be ₵21250. At $\alpha = 0.05$, what will be the conclusion of the sales manager? (4½ marks)
- d) The Quality Assurance Director of an educational institution wants to verify whether the mean grade point average (GPA) of some 100 students who did late registration is below 2.36. He randomly checked the GPA of 30 students and computed the mean GPA as 2.66. The standard deviation of the GPA of the 30 students is 0.98. At $\alpha = 0.01$, is there enough evidence to support the Director's claim? (4½ marks)

SECTION B

ANSWER ANY THREE QUESTIONS FROM THIS SECTION EACH QUESTION IN THIS SECTION CARRIES 15 MARKS

QUESTION TWO

The management of a Graduate Alumni Co-operative Credit Union wishes to invest ₵50000 into a portfolio comprising three investments earning a simple interest of 6%, 7% and 10% per annum respectively in order to receive an annual interest income totaling ₵3840. The management wants the interest earned from the investment earning 10% to be ₵540 less than the total interest from the investments earning 6% and 7% per annum. Suppose the above problem is represented by the following system of equations:

$$r + s + t = 50000$$

$$6r + 7s + 10t = 384000$$

$$10t - 7s - 6r = -54000$$

where r , s and t are the respective amounts to be invested at 6%, 7% and 10% per annum.

- Write the system of linear equations in a matrix form.
- Using matrix algebra, find how much should be invested at each rate to meet the expectation of management of the Union.

QUESTION THREE

A small mineral refinery firm manufactures two jewelry products namely necklaces and bracelets from gold, platinum and titanium. The refinery requires 9 and 3 ounces of gold to produce each necklace and bracelet respectively. In addition, each necklace requires 5 and 6 ounces of titanium and platinum respectively. The factory has a total of 240 and 275 ounces of platinum and titanium respectively. To make each bracelet stronger, the factory adds 4 and 5 ounces of platinum and titanium respectively, to the ounces of gold required. The firm has procured only 270 ounces of gold from a gold trader. In all, it costs the firm \$415 and \$455 to produce a bracelet and a necklace respectively, while a necklace is sold at \$560 and a bracelet at \$515. The firm wants to determine the number of necklaces and bracelets to make in order to attain the highest profit.

- (i) Formulate a linear programming model for the above production problem.
- (ii) Using graphical analysis, find how many necklaces and bracelets the firm should produce and the largest profit from this production.
- (iii) Determine the scarce resources of the firm.

QUESTION FOUR

An Agro-Company produces maize for supply to various outlets in Ghana. The fixed cost expenditure of the firm is ₵6 while its demand function as $P = \frac{Q^2}{3} - 5Q + 26$ where P is the price and Q is the quantity of maize produced. The addition to total cost as a result of producing one extra acre of maize is given by the expression $MC = 56 - 23Q + 2Q^2$. All figures associated with the company are measured in ten thousands of Ghana Cedis. Find

- (i) The firm's total cost function and total revenue function.
- (ii) The minimum total cost and the output level at which the firm minimizes total cost.
- (iii) The maximum profit and the output level at which the firm maximizes profit.

QUESTION FIVE

A monopolistic company has to spend exactly \$3000 as total cost for the production of a number of Basic Science and Mathematics textbooks for three schools. The Basic Science textbook sells at $P_1 = 455 - Q_1 - Q_2$ and the Mathematics textbook at $P_2 = 910 - Q_1 - 4Q_2$ where P_1 and P_2 denote the prices; Q_1 and Q_2 denote the number of Basic Science and Mathematics textbooks produced respectively. The joint total cost of producing these textbooks is given as $TC = 5Q_1 + 10Q_2$.

- (a) Find the maximum profit the producer can make.
- (b) Estimate the new optimal profit if the company decides to reduce the total cost by \$50. (Assume that second order conditions are satisfied)

QUESTION SIX

(a) The University Librarian wants to confirm whether the proportion of all undergraduate UPSA students who accessed the Alabi Library during the revision week is higher than 65%. On the last day of the revision week, he administered 200 questionnaires at the entrance of the University to undergraduate students and found that 120 of them have accessed the library in the revision week. At $\alpha=0.05$, is there enough evidence to confirm the claim?

(b) The Finance Manager of a company wants to explain the variation in profit, measured in thousands of Ghana Cedis, to shareholders of a company using regression analysis in Microsoft Excel. The regression output below was produced by the Finance Manager where Profit is the dependent variable while Sales, in thousands of Ghana Cedis, Size of their shops measured in square metres, GDP in billions of Ghana Cedis, Capital and Labour in units are the independent variables. Use the regression output to answer the questions below.

Regression Statistics	
R Square	0.8496
Adjusted R Square	0.8409
Observations	31

ANOVA					
	df	SS	MS	F	Significance F
Regression	5	47.4804705	9.496094	4.030386	0.0009
Residual	25	58.9031301	2.356125		
Total	30	106.3836006			

Dep. Variable: Profit	Coefficients	Standard Error	t-stat	P-value	Lower 95%	Upper 95%
Intercept	2.704305	0.440352	6.14	0.000	1.824332	3.584279
Sales	0.492330	0.124566	3.95	0.000	0.243404	0.741256
Capital	0.609715	0.475821	1.28	0.201	-0.325817	1.545247
ln(Labour)	-19.888750	9.258442	2.15	0.032	-38.092180	-1.685311
Size	0.415252	0.104557	3.97	0.000	0.206437	0.624068
GDP	0.012164	0.016792	0.72	0.469	-0.045180	0.020852

- Provide an interpretation for the coefficients of Sales, Capital and Labour respectively.
- At $\alpha=0.01$, test whether the coefficient of Sales is statistically different from zero.
- At $\alpha=0.05$, test whether the coefficient of Capital is statistically significant.
- Explain the overall explanatory power of the regression model estimated.

Formula Sheet and Critical Values

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}; \quad z = \frac{\hat{p} - p}{\sqrt{pq/n}}; \quad t = \frac{\bar{x} - \mu}{s/\sqrt{n}}; \quad t = \frac{\hat{p} - p}{\sqrt{1-r^2}}; \quad A_t = P\left(1 + \frac{r}{m}\right)^{mt} \quad \text{where } t \text{ is in years.}$$

Summary of Critical Values Tables for z-test and t-test.

z-test Critical Values				t-test Critical Values				
α	Right-tail	Left-tail	Two-tails	α	df	Right-tail	Left-tail	Two-tails
0.01	2.33	-2.33	± 2.58	0.01	28	2.467	-2.467	± 2.763
0.05	1.65	-1.65	± 1.96	0.05	28	1.701	-1.701	± 2.048
0.10	1.28	-1.28	± 1.64	0.10	28	1.313	-1.313	± 1.701

t-test Critical Values				t-test Critical Values				
α	Right-tail	Left-tail	Two-tails	α	df	Right-tail	Left-tail	Two-tails
0.01	2.462	-2.462	± 2.756	0.01	30	2.457	-2.457	± 2.750
0.05	1.699	-1.699	± 2.045	0.05	30	1.697	-1.697	± 2.042
0.10	1.311	-1.311	± 1.699	0.1	30	1.310	-1.310	± 1.697

*****END OF PAPER*****

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